Supplementary Information for

Conditional validity of quantum event classifiers under collider systematics and quantum estimation uncertainty

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Scope of statistical guarantees. Theorem 1's anytime-valid error control applies to each fixed I2 label-stream claim, conditional on the frozen finite audit population under the declared with-replacement sampling protocol. I3 profile-likelihood and auxiliary-information procedures are separately coverage-gated. The CMS and other empirical ledgers use their own calibrated tests and information constraints and do not inherit Theorem 1 automatically.

This supplement gives the full algebra behind the formal results, the registered-falsifier and correction trail, and extended numerical tables. Every numerical table is a transcription or an explicitly identified summary of the immutable JSON artifact named in its caption. Superseded first-run tables remain in the repository under `*_v1_*.json`.

S1 Formal details

S1.1 Exact weighted anytime-valid certification

Condition on a frozen finite audit population and a scalar $w_{\max}$ fixed before the random audit order. For IID with-replacement labeled draws (c_i, u_i), with $c_i \in \{0,1\}$, $u_i \in [0, w_{\max}]$, $E[u] > 0$, and $R = E[uc]/E[u]$, fix $\tau \in [0,1]$ before observing the audit label stream and define

$$Z_i(\tau) = \frac{u_i(c_i - \tau) + \tau w_{\max}}{w_{\max}}.$$

Because $u_i(c_i - \tau)$ lies between $-\tau w_{\max}$ and $(1-\tau)w_{\max}$, $Z_i(\tau) \in [0,1]$. Moreover,

$$E[Z(\tau)] \geq \tau \iff E[u(c-\tau)] \geq 0 \iff E[uc] \geq \tau E[u] \iff R \geq \tau.$$

Equivalently,

$$E[Z(\tau)] - \tau = \frac{E[u]}{w_{\max}}(R - \tau).$$

The reduction is therefore exact. If (L_n, U_n) is any level $1-\alpha$ confidence sequence for the mean of Z, then on its time-uniform coverage event $L_n \geq \tau$ implies $R \geq \tau$ at every n. Thus

$$P\{\exists n : \text{SUPPORTED and } R < \tau\} \leq \alpha$$

under arbitrary stopping; symmetrically, $P\{\exists n : \text{REFUTED and } R \geq \tau\} \leq \alpha$. Both kinds of incorrect decisive verdict are contained in the same two-sided CS coverage failure, so their union for this fixed claim is also bounded by α. Setting $u \equiv 1$ and $w_{\max} = 1$ recovers the unweighted construction. Weighting changes the variance and hence the stopping time, but adds no validity slack. Labeling reveals (y_i, w_i) and therefore $u_i = w_i$, $w_i 1[y_i = 1]$, or $w_i 1[y_i = 0]$ as appropriate; event-wise weights remain unavailable to I1. The data-curation layer supplies only the single global scalar bound before sampling, not row weights or class labels; this does not enlarge I1 with event-level information. Both classes have positive archived weight mass, establishing $E[u] > 0$ for the class-conditional cases. The multiplier 2.05 covers the largest admissible compound scale $2.0 \times 1.01 = 2.02$.

This guarantee is per fixed claim and simultaneous only over its labeling times/stopping rules. It is not FWER or simultaneous control over thresholds, models, or environments; selecting τ or a claim after labels requires a separate multiplicity construction. Time-uniformity also does not validate arbitrary adaptive row selection. E07 instead uses a score-based proposal fixed before labels and bounded importance weights. The contribution claimed here is this exact fixed-τ ratio reduction and its integration with the information-conditional auditor, not priority over importance-weighted, self-normalized or adaptive-data confidence sequences; the directly relevant off-policy literature is cited in the main manuscript.

S1.2 Information-set boundary for weight-only nuisances

I1 is the fixed-size or count-conditioned feature experiment $\mathcal{O}_1^{(n)} = (X_1, \ldots, X_n) \mid N = n$; it excludes N, exposure, yields and event weights. If a weight-only nuisance satisfies

$$P_\theta(X_1, \ldots, X_n \mid N = n) = P_0(X_1, \ldots, X_n \mid N = n),$$

every I1 statistic has the same sampling law. At I2 the simulated protocol reveals a binary signal/background label and nominal weight for each queried draw. The label does not reveal the background process category. If the joint law of $(X, Y, w^{(0)})$ remains unchanged while the nuisance-dependent category or true weight $w^{(\theta)}$ is hidden, the same indistinguishability result holds. A level-α test then has power equal to its actual size and hence at most α; under the fail-closed policy, affected rate and true-weighted claims remain UNRESOLVED.

This statement is deliberately limited to the declared experiment. The full observable collider experiment may instead contain $P_\theta(N, X_1, \ldots, X_N)$. At I3 a control-region count $N \sim \mathrm{Poisson}(\lambda(\theta))$ with $\lambda(\theta) \neq \lambda(0)$ changes that law and restores nontrivial power. The restoration is quantitative rather than absolute: template variance

$$\sigma_c^2 = \sum_g (\mathrm{relerr}_g \, \lambda_g)^2$$

sets the resolution scale. This is why the template-statistics-aware E14 fit identifies the ttbar rate only to approximately $\pm 10\%$ and leaves tighter claims unresolved.

S1.3 Random deployments and verdict stability

Let ω denote the finite-shot or hardware realization and $\widetilde{f}_\omega$ the resulting refit, recalibration, and threshold. If, conditional on ω, the audit stream remains IID and disjoint from training, calibration, and threshold selection, and the audit controls false certification,

$$P(\text{false certification} \mid \omega) \leq \alpha,$$

then the tower property gives marginal control for both the deployment-relative and ideal-anchored claim classes:

$$P(\text{false certification}) = E_\omega[P(\text{false certification} \mid \omega)] \leq \alpha.$$

For signed target and source movements ΔM_T and ΔM_S, the exact margin movements are ΔM_T for the ideal-anchored claim and $\Delta M_T - \Delta M_S$ for the deployment-relative claim. Therefore $|m^\star| > |\Delta M_T|$ or, respectively, $|m^\star| > |\Delta M_T - \Delta M_S|$ is sufficient to preserve the truth-sign; failure of a sufficient condition gives no conclusion. For either claim class, the corresponding sign-stability condition plus coverage of both CSs plus resolution of both audits implies the same decisive verdict. Conditional on a realization satisfying stability, two level-α audits obey

$$P\{V^\star \neq V(\omega), R^\star \cap R(\omega) \mid \omega\} \leq 2\alpha$$

by a union bound. Running each audit at $\alpha/2$ gives joint level-α control; no sharper dependence-based result is used. Coverage and resolution alone are insufficient: $m^\star = 0.05$, $\Delta M_T = -0.10$ yields opposite truth signs and opposite perfectly covered verdicts. Failure does not force a flip either: $\Delta M_T = +0.10$ violates the inequality but preserves the sign. For weighted claims the metric-scale margin is mapped to the certification scale by $E[u]/w_{\max}$. Common-mode movement cancels from the deployment-relative margin but not from the ideal-anchored margin. Although the original E16 summary archived only ΔM_S, the final deterministic replay reconstructs ΔM_T and $\Delta M_T - \Delta M_S$ from the frozen source/target rows and the exact replayed raw and PSD-repaired deployments. It therefore instantiates both sufficient conditions without new randomness or QPU jobs; the resulting 7,200-case evaluation and its conservative interpretation are reported in Section S3.3.1 and Table S8.

S1.4 Pre-split component allocation for weighted balanced accuracy

For $\mathrm{BA}_w = (\mathrm{TPR}_w + \mathrm{TNR}_w)/2 \geq \tau_{\mathrm{BA}}$, the mathematical construction can predeclare component thresholds τ_1, τ_2 satisfying $(\tau_1 + \tau_2)/2 = \tau_{\mathrm{BA}}$ and error levels $\alpha_1 + \alpha_2 = \alpha$. Each component uses the exact one-sample reduction with a valid class-specific bound. The BA claim is SUPPORTED only if both components are supported, REFUTED only if both are refuted, and UNRESOLVED otherwise. If BA is false, at least one component claim is false; certifying it requires a coverage violation. A union bound therefore controls false certification at α. The symmetric argument controls false refutation. In the executed E13v2 battery, however, the class maxima were calculated using labels from the complete frozen population. It is therefore an oracle/benchmark diagnostic rather than an operational I2 guarantee. Table S11 shows that even this favorable oracle construction is information-limited by the benchmark's weighted signal fraction; successful TNR resolutions are reported only in that diagnostic sense.

S2 Registered falsifiers and audit trail

The count below is by *unique registered falsifier arm*. The E15 calibration gate is one arm, although it rejected two successive invalid implementations before the valid grid was allowed to run. This convention reconciles the nine-arm tally with the two gate interventions.

Table S1: All nine registered falsifier arms that fired. Frozen conditions and consequences are recorded in `docs/experiment_registry.md`.

Arm	Frozen condition	Observed firing	Consequence obeyed
E02R	TES sign pattern fails multi-seed replication	Up-shift monotonicity in only 2/5 seeds	Down-shift retained provisionally within-world; E17 later withdrew the directional cross-world claim.
E12(e)	Two flagship decoupling cells fail to reproduce	Diboson flagship cells stayed nominal-like in the fresh world	Normalization-collapse statement scoped to signal-region composition; failing registered cells kept.
E14 v1	Control-region interval coverage below nominal	Pure-Poisson fit gave ttbar bias +0.07 and zero coverage	Run blocked; template-statistics likelihood registered, then rerun as v2.
E15 gate	Nominal L2 coverage outside 0.6827 ± 0.02	Conditional-ensemble and numerical-conditioning implementations each failed	Both grids blocked; ensemble semantics and optimizer fixed before any shifted claim.
E17(b)	Degradation signs fail across additional worlds	One world improved QK under combo by 0.009; another under TES-down by 0.005	Cross-world degradation claim withdrawn.
E08v2(a)	Corrected counting coverage outside the nominal band	One-draw corrected coverage 0.9916–1.000	Arm blocked and bounded multi-draw E08v3 registered.
E08v2(b)	Independent-template flagship L2 coverage < 0.633	Coverage 0.000 versus shared 0.7188	"Profiling restores validity" made explicitly shared-simulation-conditional.

Arm	Frozen condition	Observed firing	Consequence obeyed
E08v3(a)	Marginal symmetric-BB coverage outside 0.6827 ± 0.06 for at least two gated models	All three gated models overcovered, 0.7513–0.8284	Calibration-restoration claim withdrawn; conservative direction reported.
E13v2(b)	All true benchmark BA claims unresolved by 5,000 labels	TPR component 0/200 resolutions in every cell	BA path published as information-limited; TNR success and mechanism reported separately.

Table S2: Audit-trail summary. The full findings and dispositions are in `docs/audits/`.

Audit	Scope	Material effect
Pre-campaign, 2026-08-11	All E00–E11 code, tables, splits, and governance	Seven design constraints frozen before E12–E16: archived row indices, role separation, post-selection sensor validation, an independent floor, independent quantum-noise streams, a typed deployment snapshot, and weighted estimands. No prior headline invalidated.
Post-campaign, 2026-08-11	E12–E16, E04v3, E11v2; 172 registry values and about 178 manuscript claims	E13 nominal-weight estimand fixed and rerun; E15 gate rejected two implementations; E16 dual accounting added; E12 normalization reading corrected; all superseded tables retained.
Pre-submission, 2026-08-12	Extension runners, figure scripts, and line-by-line formal proofs	Two HIGH findings fixed: JES tracking scoped and disclosed; E19 weighted estimand corrected and rerun. Proposition 4 gained its coverage-event and scale-bridge qualifiers. Eight LOW findings were fixed, completed, disclosed, or documented.
F8.2, 2026-08-12	Final LaTeX, supplement, number trace, semantic equivalence, rendered PDFs	E08v3 bias range corrected from +6.1 to the archived +7.5 endpoint; the nine-arm counting convention made explicit.
Final mathematical/editorial, 2026-08-12	Proposition 2, Theorem 1, Proposition 4, Contributions 2/3, E17, figures, and three independent adversarial reviews	Claims were restricted to their exact mathematical and evidential scope: per-claim rather than family-wise control; finite-population-conditional weight bounds; conditional Proposition 4 with empirical flips separated; joint nuisance-representability/template-quality limits; no new experiment or result.
Senior-author pre-submission, 2026-08-13	Observable experiments, Proposition 4, weighted-inference literature, E16 deployment unit, C2 narrative, CMS wording and concision	Added the missing sign-stability premise and counterexample; separated feature-conditioned evidence from counts; reanalyzed all 30 existing E16 deployments without new runs; retired IID environment p-values; updated related-manuscript status and reduced audit-history prose in the main paper.
Final pre-submission patch, 2026-08-31	Primary literature, novelty claims, abstract hierarchy, methods, supplement, release metadata and rendered artifacts	Added the closest competitive work, removed priority language, made Contribution 3 deployment semantics primary, expanded protocol detail, and synchronized editorial/release records. No experiment, dataset, model, configuration or primary result changed.
Final technical PSD audit, 2026-08-31	The 30 frozen E16 deployments, LIBSVM semantics, hardware-entry provenance and sealed-role wording	Deterministically replayed every raw deployment, measured the complete training-Gram spectra, applied one declared minimum-diagonal-loading sensitivity, and classified the result PSD-SENSITIVE-BUT-SCOPED. No new Gram realization, seed, sample or QPU job was generated.

S3 Extended numerical results

S3.1 Inference sensitivity (E15)

Table S3 reports the local slope at the nominal point, averaged over the five injected μ values. L1 is the counting estimator, L2 the complete profile, and L3 the profile with the shifted family omitted. Only the three models that passed the nominal L2 gate are shown. Units are μ per nuisance unit: per standard deviation for TES and JES, per GeV for soft-MET, and per unit scale for the normalization families.

The archived initial global optimizer-success flag is conservative and falls as low as 0.382. Analytic gradients, multi-start fitting, a monotone lower-minimum safeguard, and the nominal coverage gate support the reported solutions but do not prove global optimization in every shifted cell.

Table S3: E15 local sensitivity and nuisance tracking. Source: `results/tables/E15_sensitivity.json`.

Family	Model	L1 slope	L2 slope	L3 slope	L2 tracking
TES	QK-SVC	9.8095	0.4092	7.1213	0.9970
TES	RBF-SVC	10.3065	0.2269	8.6641	0.9986
TES	XGBoost	5.4959	0.1031	4.7948	0.9938
JES	QK-SVC	3.6530	-0.2033	5.1238	0.9813
JES	RBF-SVC	2.7528	-0.1096	5.7069	0.9901
JES	XGBoost	1.7328	0.0121	1.1957	0.7116
soft-MET	QK-SVC	-0.0812	-0.2126	6.5142	0.2497
soft-MET	RBF-SVC	0.7731	-0.0496	3.0473	0.2511
soft-MET	XGBoost	0.7305	-0.4605	-1.5627	0.5001
ttbar scale	QK-SVC	49.7510	0.5175	-0.7125	1.0040
ttbar scale	RBF-SVC	54.0570	-1.1795	-4.0830	0.9915
ttbar scale	XGBoost	12.7935	-0.0220	9.0305	0.9895
diboson scale	QK-SVC	4.7626	0.0834	0.5676	0.9900
diboson scale	RBF-SVC	4.3356	0.2769	1.0303	0.9869
diboson scale	XGBoost	0.9981	-0.0065	0.4824	1.0006
background scale	QK-SVC	270.1200	-30.9900	222.4700	1.0000
background scale	RBF-SVC	397.2600	34.5600	823.4100	0.9800
background scale	XGBoost	101.8800	-5.3900	-3.6800	1.0000

S3.2 Fresh-world validity and stopping-time efficiency

Table S4: Fresh-world validity replication. Source: `results/tables/E19_fresh_world_validity.json`; the false-refutation counts documented in the registered status agree with the rates stored in the table.

Accounting	False certification	False refutation	$\alpha + 3\sigma$ boundary
Unweighted, non-vetoed	11/3,060 = 0.359%	1/11,980 = 0.008%	6.182%
Unweighted, all streams	11/7,700 = 0.143%	1/11,980 = 0.008%	6.182%
Weighted, nominal weights	6/7,980 = 0.075%	4/11,700 = 0.034%	5.732%

All 12 archived model–environment score comparisons were byte-identical before auditing. The weighted bound was $w_{\max} = 7.22421 \times 2.05 = 14.80963$. The corrected nominal-weight estimand is invariant across all 12 weight-only environments.
These correlated Monte-Carlo rates are implementation stress tests, not a proof of the theorem or family-wise error control across the claim grid.

Table S5: Audit-label draw-budget price relative to the Wald information yardstick. Sampling is with replacement, so n^* is a labeled-observation budget rather than a unique-event label cost. Source: `results/tables/E06_nstar_efficiency.json`.

Subset	Cells	Q25	Median	Q75
All resolved cells	518	1.56	2.07	2.97
Margin $[0.01, 0.02)$	95	2.82	3.35	3.85
Margin $[0.02, 0.04)$	95	2.62	3.19	3.66
Margin $[0.04, 0.08)$	164	1.63	1.98	2.38
Margin ≥ 0.08	164	1.25	1.46	1.72
Certification direction	487	1.54	1.99	2.78
Refutation direction	31	3.12	3.34	3.65

S3.3 Quantum-estimation diagnostics (E16)

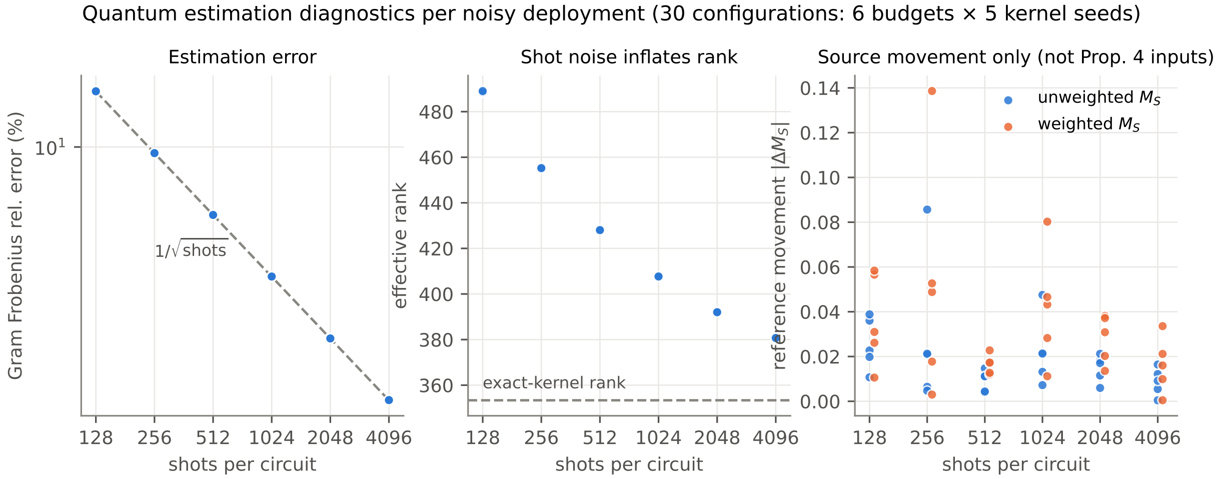

Figure S16: Per-configuration diagnostics for all 30 noisy deployments: relative Frobenius error, effective-rank inflation, and source-reference movement $|\Delta M_S|$. The primary artifact plotted only source movement; the final deterministic raw/PSD replay reconstructs ΔM_T and $\Delta M_T - \Delta M_S$ and instantiates Proposition 4 in Table S8. Sources: `results/tables/E16_quantum_uncertainty.json` and `results/tables/E16_proposition4_instantiation.json`.

The primary replication unit is the independently sampled noisy-kernel realization, five per shot budget. Within a deployment, 200 far, 170 moderate and 230 near claims share the Gram matrix, refit, calibration, threshold and audit streams; their count is not an IID sample size. Audit streams are paired across deployments, so comparisons also share common random numbers. The far ideal-anchored deployment means are 20.8, 17.7, 0.9, 11.9, 5.8 and 0.4%, so the intermediate sequence is non-monotonic. At the endpoints the sample SDs are 19.9 and 0.65%, ranges are 0–44 and 0–1.5%, and leave-one-deployment-out means are 15–26 and 0.13–0.50%. These are descriptive dispersion and sensitivity summaries, not population confidence intervals. Five deployments per budget permit only a descriptive characterization of the observed realizations; no population trend or equivalence claim is made. All far-margin deployment-relative claims in this evaluated grid are true and remain SUPPORTED in every raw and PSD-repaired realization. Consequently, the zero far-margin flip rate demonstrates stability of clearly true/supported claims only; this grid contains no false far-margin deployment-relative claim with which to assess refutation stability.

Table S6: E16 results by noisy-kernel deployment. Rates are percentages of the correlated claims within that deployment, not independent replication. "Fixed" is ideal-anchored and "own" is deployment-relative. Source: `results/tables/E16_quantum_uncertainty.json`; derived deployment summary: `results/tables/E16_deployment_level.json`.

Shots	Seed	Frob. (%)	AUC	Far flips		Moderate flips		Near abst.
				fixed	own	fixed	own	own
128	1	13.64	0.8336	0.0	0.0	52.3	14.7	95.7
128	2	13.69	0.8311	11.5	0.0	74.7	15.9	94.8
128	3	13.66	0.8338	40.0	0.0	80.0	14.1	93.9
128	4	13.66	0.8377	44.0	0.0	78.8	18.2	93.0
128	5	13.67	0.8384	8.5	0.0	71.2	15.9	92.6
256	1	9.67	0.8361	0.0	0.0	50.0	18.2	92.2
256	2	9.66	0.8331	0.0	0.0	34.1	11.8	93.9
256	3	9.67	0.8278	0.0	0.0	50.0	12.3	92.2
256	4	9.65	0.8364	0.0	0.0	32.9	16.5	90.4
256	5	9.66	0.8436	88.5	0.0	80.0	21.8	95.7
512	1	6.85	0.8307	2.0	0.0	62.4	9.4	96.1
512	2	6.84	0.8394	0.0	0.0	30.0	10.6	91.3
512	3	6.83	0.8480	1.0	0.0	57.6	12.3	90.9
512	4	6.83	0.8486	0.0	0.0	30.6	10.6	93.5
512	5	6.83	0.8478	1.5	0.0	57.6	14.7	90.9
1024	1	4.83	0.8385	0.0	0.0	49.4	15.3	90.9
1024	2	4.83	0.8338	0.0	0.0	51.2	12.9	90.9
1024	3	4.84	0.8381	0.0	0.0	47.1	11.8	93.9
1024	4	4.83	0.8381	59.0	0.0	80.0	15.9	93.0
1024	5	4.84	0.8385	0.5	0.0	45.3	10.0	91.3
2048	1	3.42	0.8366	0.0	0.0	32.9	10.0	94.8
2048	2	3.42	0.8460	0.0	0.0	49.4	10.0	93.0
2048	3	3.42	0.8469	3.0	0.0	61.2	12.9	90.4
2048	4	3.42	0.8496	14.0	0.0	73.5	11.2	96.5
2048	5	3.42	0.8428	12.0	0.0	74.7	11.2	94.8
4096	1	2.42	0.8409	0.0	0.0	39.4	4.7	93.0
4096	2	2.42	0.8444	0.0	0.0	3.5	5.3	91.7
4096	3	2.42	0.8357	1.5	0.0	56.5	10.6	94.8
4096	4	2.42	0.8373	0.0	0.0	48.8	8.2	92.2
4096	5	2.42	0.8400	0.5	0.0	53.5	4.7	93.5

Table S7: Post-hoc E16 PSD sensitivity prompted by the final technical audit. Each row summarizes five deterministically replayed frozen deployments. Flip entries are mean far-margin percentages relative to the ideal exact-kernel deployment; "dep." uses each realization's refrozen source reference and "ideal" uses the fixed ideal reference. Minimum diagonal loading preserves every off-diagonal and cross-Gram estimate. Source: `results/tables/E16_psd_sensitivity.json`.

Shots	n	med. $\lambda_{\min}$	worst $\lambda_{\min}$	med. n_-	max n_-	dep. raw	dep. PSD	ideal raw	ideal PSD
128	5	-1.930	-1.939	580	580	0.0	0.0	20.8	31.5
256	5	-1.325	-1.333	531	532	0.0	0.0	17.7	40.7
512	5	-0.936	-0.966	489	490	0.0	0.0	0.9	25.1
1024	5	-0.655	-0.657	449	450	0.0	0.0	11.9	3.0
2048	5	-0.448	-0.457	412	413	0.0	0.0	5.8	6.9
4096	5	-0.313	-0.319	379	380	0.0	0.0	0.4	0.3

All 30 raw 2000×2000 training Grams are indefinite under the declared negative-mode tolerance $10^{-10}\max(1, |\lambda_{\max}|)$. Across budgets, the median negative spectral-mass fraction falls from 13.59% to 2.14%, while $\lambda_{\max}$ remains approximately 201.2; the positive-only condition number ranges from 8.8×10^4 to 6.4×10^7 and is descriptive only because the raw matrices are indefinite. The historical finite-shot pipeline fits an SVC to each realized quantum-similarity matrix. Entry-wise finite-shot estimation does not preserve positive semidefiniteness, so this RAW-INDEFINITE fitted object is not interpreted as the standard convex RKHS SVM associated with a PSD kernel. The material far-margin changes are confined to the ideal-anchored magnitude: they are substantial at 128–1024 shots, small at 2048 and negligible at 4096; the deployment-relative far rate is unchanged at every budget. LIBSVM can and did return a numerical solution when supplied each precomputed raw matrix; indefiniteness alone also does not make its numerical deployment meaningless.

The historical RAW-INDEFINITE pipeline remains primary. For the declared sensitivity, each training Gram is converted in float64 as $K_{\mathrm{PSD}} = K_{\mathrm{raw}} + \delta I$, where $\delta = \max(0, -\lambda_{\min} + \epsilon)$ and $\epsilon = 10^{-10} \max(1, |\lambda_{\max}|)$ (approximately 2.01×10^{-8} here). This is the minimum predeclared diagonal load up to the numerical margin; it preserves all off-diagonal finite-shot estimates. The loaded training block restores a convex precomputed-SVM training problem, but it is a post-hoc regularized similarity matrix, not a normalized fidelity Gram: the diagonal changes, source/target cross-Grams remain unchanged, and no global Mercer extension is claimed. Each repaired deployment is then refitted, Platt-recalibrated, threshold-refrozen and audited on exactly the same roles, claim grids and paired audit streams. No Gram, sample, seed or QPU job is newly drawn.

The PSD-repaired nominal-AUC means differ from raw by +0.0051 to +0.0137 across budgets; these changes are a sensitivity outcome, not a repair-selection or performance claim. Raw-versus-repaired deployment-relative verdict-change rates are zero in the far stratum, at most 19.9% in moderate and 7.6% in near; ideal-anchored compositions move more strongly. Nevertheless, the observed support stability of the evaluated true far-margin deployment-relative claims, ideal-anchored heterogeneity, non-monotonic budget sequence and endpoint reduction all survive. The exact deployment metrics, thresholds, target metrics, abstention, verdict compositions and transitions are archived in the source JSON. The audit verdict is therefore PSD-SENSITIVE-BUT-SCOPED: Contribution 3's qualitative statement survives, whereas its raw ideal-anchored percentages must not be treated as repair-invariant.

S3.3.1 Deterministic instantiation of Proposition 4

For the unweighted family, M is mean correctness on the frozen source or target rows; for the weighted family it is raw-physical-weighted mean correctness on those rows. For each deployment ω, the replay computes $\Delta M_S = M_S(\tilde{f}_\omega) - M_S(f^\star)$ and $\Delta M_T = M_T(\tilde{f}_\omega) - M_T(f^\star)$. The evaluated movement is $\Delta M_T - \Delta M_S$ for deployment-relative claims and ΔM_T for ideal-anchored claims. Every reconstructed threshold is interior to $[0, 1]$, and both exact margin identities have zero numerical residual, so all 7,200 condition cells are evaluable.

Table S8: Proposition 4 instantiation. "H" and "F" mean that the strict sufficient condition holds or fails; entries are verdict flip/no-flip counts over the ten paired audit streams per condition cell. Cells and streams within a deployment are correlated; the descriptive unit remains the deployment. Source: `results/tables/E16_proposition4_instantiation.json`.

Cut	Condition cells	H (%)	H: flip/no flip	F: flip/no flip
Overall	7,200	68.7	4,554 / 44,876	13,637 / 8,933
RAW-INDEFINITE	3,600	70.3	2,260 / 23,060	6,247 / 4,433
PSD-repaired	3,600	67.0	2,294 / 21,816	7,390 / 4,500
Deployment-relative	3,600	92.8	2,127 / 31,283	62 / 2,528
Ideal-anchored	3,600	44.5	2,427 / 13,593	13,575 / 6,405
Far	2,400	95.3	683 / 22,197	967 / 153
Moderate	2,040	70.0	2,955 / 11,325	4,112 / 2,008
Near	2,760	44.5	916 / 11,354	8,558 / 6,772

The inequality holds in 4,943/7,200 cells and preserves the truth sign in all 4,943, while 1,188 of the 2,257 failing cells also retain their sign. At audit level, ternary verdict flips occur in 9.2% of streams when the condition holds and 60.4% when it fails. Two condition-holding streams produce opposite resolved verdicts; Proposition 4 bounds such events through confidence-sequence coverage and does not assert deterministic verdict equality from the margin inequality alone. The condition is therefore informative but conservative: it is most useful for deployment-relative far/moderate cells and least often satisfied for near ideal-anchored cells; it is sufficient rather than necessary. This classification is INFORMATIVELY INSTANTIATED, not an empirical general law or a set of independent claim replications.

S3.4 Independent-MC split studies (E08v3)

Table S9: Marginal nominal counting coverage over 400 independent splits. Source: `results/tables/E08v3_multidraw.json`.

Model	Shared truth	Indep. naive	Indep. BB	Indep. BB symmetric	Stored prediction
QK-SVC	0.6828	0.0691	0.6501	0.8113	0.5220
RBF-SVC	0.6827	0.0860	0.6605	0.8284	0.5220
XGBoost (A)	0.6830	0.0673	0.5650	0.7513	0.5216
XGBoost (B)	0.6823	0.0645	0.5458	0.7033	0.5164

Table S10: Independent-template fixed-profile results in the two registered E08v3 cells across 10 splits, averaged over the frozen μ grid within each split. This does not test MC-statistics-aware profiling in general. Source: `results/tables/E08v3_multidraw.json`.

Cell	Coverage range	Draws < 0.633	Bias range	Shared reference
TES = 0.98, XGBoost	0.000–0.238	10/10	-6.600–$+4.050$	0.7188
Nominal, XGBoost	0.000–0.359	10/10	-6.557–$+7.496$	gate-passing in E15

S3.5 Weighted balanced-accuracy allocation (E13v2)

Table S11: E13v2 oracle component-allocation battery, 200 replications per cell at $n_{\max} = 5{,}000$. Class bounds use full-population labels, so this is not an operational I2 guarantee. Entries are correct resolutions; all false-certification counts are zero. Source: `results/tables/E13v2_baw_allocation.json`.

Population	Claim	BA pre-split	TPR component	TNR component	v1 BA	median n^*
Synthetic	$m = 0.02$, true	0/200	9/200	36/200	0/200	–
Synthetic	$m = 0.02$, false	0/200	3/200	36/200	0/200	–
Synthetic	$m = 0.05$, true	83/200	83/200	199/200	0/200	3,489
Synthetic	$m = 0.05$, false	87/200	87/200	200/200	0/200	3,370
Benchmark	$m = 0.02$, true	0/200	0/200	67/200	0/200	–
Benchmark	$m = 0.02$, false	0/200	0/200	64/200	0/200	–
Benchmark	$m = 0.05$, true	0/200	0/200	200/200	0/200	–
Benchmark	$m = 0.05$, false	0/200	0/200	200/200	0/200	–

Table S12: Mechanism diagnostics for E13v2 at $n = 5{,}000$.

Diagnostic	Synthetic control	Benchmark physics weights
Weighted signal fraction	0.301459	0.000966
Positive-class ESS ratio	0.226298	0.001106
$w^{(+)}_{\max}$	4.58547	5.98645
$w^{(-)}_{\max}$	9.40021	14.89144
TPR Z-margin at $m = 0.05$	0.01130253	0.00002827
TPR CS radius	0.011776	0.001806
v1 BA radius	0.168819	0.288855

In the benchmark, the TPR radius-to-margin ratio is about 64 at 5,000 labeled draws. A quadratic information-scaling extrapolation therefore gives $n^* \approx 5{,}000 \times 64^2 \simeq 2 \times 10^7$ uniform labeled draws at margin 0.05. This is a diagnostic extrapolation, not a new Monte Carlo run.

S4 Computational, inference and hardware protocol details

This section collects scientific choices that are executable in repository configuration but are needed to evaluate the experiment without navigating the YAML files. It documents the frozen protocol; it does not introduce a new analysis.

S4.1 Features, data roles and hyperparameter selection

The eight inputs shared by QK-SVC and its matched RBF control are listed in Table S13. All are defined for every selected event. The choice was made from source-domain physics knowledge and frozen in D-011 on 10 August 2026, before E01. The criterion was to retain documented HiggsML discriminants while avoiding jet-dependent sentinel variables, whose fixed absent-object value could dominate an angle encoding. No target sample, target label, systematic landscape, or CMS collision data was used to select them.

Table S13: Frozen eight-feature input in circuit order. Definitions follow the FAIR Universe HiggsML data documentation.

Feature	Operational definition
`DER_mass_transverse_met_lep`	Transverse mass of the lepton and missing transverse momentum.
`DER_mass_vis`	Invariant mass of the visible hadronic-tau and lepton four-vectors.
`DER_pt_ratio_lep_had`	Ratio $p_T^{\mathrm{lep}}/p_T^{\mathrm{had}}$.
`DER_met_phi_centrality`	Azimuthal centrality of the MET vector relative to the hadronic tau and lepton.
`DER_deltar_had_lep`	ΔR separation between the hadronic tau and lepton.
`DER_pt_h`	Magnitude of the vector sum of hadronic-tau, lepton, and MET transverse momenta.
`DER_sum_pt`	Scalar sum of hadronic-tau, lepton, and all selected-jet transverse momenta.
`PRI_met`	Magnitude of missing transverse momentum.

The development draw contains 300,000 raw rows and 274,004 rows after the official nominal selection. A raw-row stratified partition gives train 109,699; `source_val` 41,128; `nominal_test` 41,116; `auditor_dev` 41,001; and sealed `final_eval` 41,060. Tier A uses a fixed stratified 2,000-event subset of train; Tier B uses all 109,699 train events. Training weights preserve physical within-class importance but equalize the two class masses and are renormalized to mean one. Evaluation always uses raw physical weights. The `source_val` role is used only for weighted Platt calibration, the weighted-balanced-accuracy threshold, and the separately declared inference bin/SR choices. `nominal_test` supplies reported nominal performance and templates; `auditor_dev` supports audit development. The full raw draw and its labels were available once when the deterministic label-stratified role indices were constructed. Sealing starts after those indices are persisted. Thereafter the loader drops `final_eval` by default; no experiment uses the explicit opt-in flag. Thus the sealed roles have not been used after sealing for model or hyperparameter selection, calibration or threshold choices, claim construction, or reported evaluation. This is a scientific-use guarantee, not a claim that their bytes were never technically loaded during split construction.

Hyperparameters were chosen by a seed-202 random search on the train role: 10 unique configurations per model family except five for the MLP, with three stratified folds and physics-weighted fold AUC. Scale-sensitive preprocessing was fitted inside each fold. The same configuration count applied to Tier A families, including QK-SVC; Tier B repeated the classical search on its own train budget. The finite search spaces were:

- linear SVC: $C \in \{0.01, 0.03, 0.1, 0.3, 1, 3, 10\}$;
- RBF SVC (28-feature and matched eight-feature versions): $C \in \{0.1, 0.3, 1, 3, 10, 30\}$ and $\gamma \in \{\text{scale}, 0.01, 0.03, 0.1, 0.3\}$;
- XGBoost: trees $\in \{200, 400, 800\}$, depth $\in \{4, 6, 8\}$, learning rate $\in \{0.03, 0.05, 0.1, 0.2\}$, subsample and column fraction each $\in \{0.7, 0.9, 1\}$, and $\lambda \in \{0.5, 1, 3\}$;
- LightGBM: trees $\in \{200, 400, 800\}$, leaves $\in \{15, 31, 63\}$, and the same learning-rate, subsample, column-fraction and λ sets as XGBoost;

- MLP: hidden widths $\in \{(32), (64, 32), (128, 64)\}$, $L_2 \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$, and learning rate $\in \{10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$;
- QK-SVC: $C \in \{0.1, 0.3, 1, 3, 10\}$, repetitions $\in \{1, 2\}$, bandwidth scale $\in \{0.25, 0.5, 0.75, 1\}$, and linear or full entanglement.

The winning parameters are in main Table 3 and the complete classical values are in `configs/frozen/frozen_deployment_v1.yaml`. They were frozen before the confirmatory world was drawn. There was no target-domain tuning.

S4.2 Counting and profile-likelihood inference

The E08 counting signal region is chosen on `source_val` from 200 score quantiles between 0.5 and 0.999 by maximizing $s/\sqrt{b}$, subject to $b \geq 50$ at the benchmark luminosity. For each environment, model and $\mu \in \{0.5, 1, 1.5, 2, 3\}$, E08 draws 2,000 Poisson pseudoexperiments from the shifted signal and background yields. Its nominal-template estimator is $\hat{\mu} = (N - b_0)/s_0$ with the reported one-standard-deviation Gaussian interval. The nominal belief and shifted truth use the same `nominal_test` rows, so this study is explicitly shared-simulation-conditional.

E15 uses a product of score-bin Poisson likelihoods. For each model, ten weighted-quantile edges are first proposed on `source_val`; low-score bins are merged until every nominal background bin has at least 25 expected events. Per-process templates for htautau, ztautau, ttbar and diboson are built from `nominal_test`. TES/JES anchors are at $-2, -1, +1, +2$ standard deviations; their per-bin deviations use a quadratic interpolation inside $|\alpha| \leq 1$ and the $1 \to 2\sigma$ linear slope outside. Soft-MET uses piecewise-linear anchors at 0, 1, 2, 3 and 5 GeV, with the three stochastic seeds averaged at nonzero anchors. Normalization nuisances act exactly and multiplicatively on ttbar, diboson and all-background yields.

TES and JES have unit-Gaussian constraints in standard-deviation units; normalization scales have Gaussian constraints with widths 0.02, 0.25 and 0.001 and the official bounds; soft-MET is flat on $[0, 5]$ GeV. Each pseudoexperiment draws its auxiliary-constraint centers around the true nuisance value, giving the declared unconditional ensemble. L2 profiles all six nuisances; L3 uses the identical likelihood but omits the nuisance family actually shifted. The 68.27% interval is $q(\mu) = -2 \log\{L(\mu, \hat{\hat{\theta}})/L(\hat{\mu}, \hat{\theta})\} \leq 1$, found with bounded L-BFGS-B fits, an analytic gradient, a second start and a monotone lower-minimum safeguard. E15 runs 500 pseudoexperiments for each model–environment–μ–level cell and 2,000 per μ for each nominal gate model before the shifted grid is interpreted.

The independent-MC studies do not repair or redefine this likelihood. E08v2 uses 500 pseudoexperiments in each of six registered profile spot-check cells. E08v3 marginalizes counting coverage over 400 independent half-split draws and uses ten profile draws with 200 pseudoexperiments per registered cell. Those studies establish the stated auxiliary-template-quality limitation; no general MC-statistics-aware profile construction is claimed.

S4.3 Quantum kernel, finite shots and hardware provenance

Training-only 0.5% and 99.5% feature quantiles map each input to $[-\pi, \pi]$, with out-of-range deployment values clipped. The frozen eight-qubit map has two repetitions and bandwidth 0.5. Each repetition applies H and $R_Z(2z_i)$ to every qubit, followed on each neighboring pair by CNOT–$R_Z(2z_i z_j)$–CNOT, where $z_i = 0.5x_i$. The fidelity kernel is $K(x, y) = |\langle\phi(x)|\phi(y)\rangle|^2$. "Exact" means statevector evaluation. "Finite shot" means the exact noiseless compute–uncompute law, sampled directly as Binomial$(S, K)/S$ at $S = 128, 256, 512, 1024, 2048, 4096$; training-Gram off-diagonals are sampled once and symmetrized, diagonals are fixed to one, and cross-Gram entries are sampled independently. No PSD projection or mitigation is applied in the historical primary pipeline. Each realization refits the SVC and repeats Platt calibration and threshold freezing. For the 2,000-event Tier-A train set, one training realization represents 1,999,000 distinct off-diagonal compute–uncompute estimates, plus $2,000 \times N$ entries for each calibration or target cross-Gram. These large finite-shot arms sample the exact Bernoulli law in software; they are not claimed as executed QPU circuits.

Two real-device jobs were run on `ibm_marrakesh` (156-qubit Heron r2). E10 job `d9t2jrvtfhrs73dtd8dg` contains 496 off-diagonal circuits for a balanced 32-event Gram at 2,048 shots. E16 job `d9tdubopdb6s73e73o20` contains 378 train-pair and 336 train–test circuits (714 total) for 28 train and 12 test events at 1,024 shots; six test events are nominal and six TES-down. Both used compute–uncompute circuits, Qiskit transpilation at optimization level 1, a fixed transpiler seed, and no error mitigation on the measured entries. The QPU-derived entries are exactly the off-diagonal training pairs and train–test cross entries reconstructed from raw counts. Training diagonals were not executed and are fixed analytically to one by the fidelity-kernel identity. Consequently, every reported hardware-Gram PSD observation refers to the constructed matrix under this unit-diagonal convention. Median transpiled depth and two-qubit count were 182 and 54; the archived QPU use was 276 s and 200 s, respectively.

Archived provenance comprises backend and job identifiers, shots, submission timestamps, circuit/pair order, feature-map parameters, aggregate transpiled depth/two-qubit counts, backend size, the calibration last-update timestamp, raw counts, row identifiers, labels, kernel blocks and E16 usage records. No explicit initial layout was fixed. Per-circuit physical layouts, transpiled circuit objects, and detailed per-qubit calibration/error tables were not archived, so the hardware evidence cannot support a calibration-conditional or layout-specific claim. It is a hardware micro-scale integration and fail-closed consistency check only; all eight-qubit experiments are classically simulable.

S4.4 CMS real-data case-study protocol

Level II uses Run2012B+C TauPlusX collision data at 11,467 pb^{-1} and the available CMS outreach MC (ggH, VBF, DY, ttbar and W+jets; no QCD sample). The full-data pass retains 126,164 selected collision events and 12,445 selected MC events. The selection requires the `IsoMu17_eta2p1_LooseIsoPFTau20` trigger, a tight isolated muon with $p_T > 20$ GeV and $|\eta| < 2.1$, a tight isolated hadronic tau with $p_T > 25$ GeV and $|\eta| < 2.3$ plus decay-mode and anti-electron/anti-muon requirements, and $\Delta R(\mu, \tau) > 0.5$. The leading passing muon and tau are used. Opposite-sign events form the analysis region and same-sign events a QCD-enriched control region. The eight features are reconstructed in the same order and with the same operational definitions as Table S13. MC weights are cross section times luminosity divided by generated events; collision events carry no class label. Models are trained on MC with a fixed 70/30 train/validation split and Level-I-frozen hyperparameters; QK-SVC uses a 2,000-event training subset. There is no collision-target tuning.

The ledger has four frozen claims. C1 asks whether event-level accuracy on collision data is within 0.05 of source accuracy and requires queryable I2 truth; it is UNRESOLVED because such labels do not exist. C2 asks whether the data/MC normalization ratio in the opposite-sign high-transverse-mass ($m_T > 70$ GeV) W-enriched control region is within 30%; E11v3 propagates MC statistics. C3 asks whether MC-to-data QK-MMD^2 exceeds the feature-only noise floor: the hardened analysis uses 200 MC–MC null draws, 20 MC–data draws and 999 permutations on unweighted 2,000-row samples, and requires both calibrations to reject. C4 asks whether the same-sign collision count exceeds the available non-QCD MC, with MC statistics in the denominator. C4 does not measure the QCD yield or establish QCD quantitatively; absent-QCD MC makes QCD a physically plausible explanation for the excess, not the certified claim.

S5 Artifact map

The main machine-readable sources for this supplement are:

- `results/tables/E15_sensitivity.json`;
- `results/tables/E19_fresh_world_validity.json`;
- `results/tables/E06_nstar.json` and `results/tables/E06_nstar_efficiency.json`;
- `results/tables/E16_quantum_uncertainty.json`;
- `results/tables/E16_deployment_level.json`;
- `results/tables/E16_psd_sensitivity.json`;
- `results/tables/E16_proposition4_instantiation.json`;
- `results/tables/E08v2_independent_mc.json` and `results/tables/E08v3_multidraw.json`;
- `results/tables/E13v2_baw_allocation.json`.

The frozen protocols and post-run dispositions are in `docs/experiment_registry.md`, `docs/decisions.md`, and the dated files under `docs/audits/`. These text records explain the design and corrections; the JSON artifacts remain the numerical source of truth.